Working with Living Soils in the Market Garden

Working with living soils: What

✅ No-Till vegetable production. Replacing mechanical tillage with Biological tillage.

✅ Working with mycorrhizal fungi to support plant health and build soil fertility

✅ Develop regenerative cropping systems, that produce high output yet, sustained or improve soils over time

Why do we farm organically ?

Grow **food** that is as **healthy** and **nutritious** for our customers

work WITH nature, healing the land. **Become real stewards**

Change the world, save it from degradation and nasty chemicals

Work with **free ecological services** of nature instead of the downward spiral of chemical farming

As a grower, how committed are you to these ideals ?

Its already hard to make a living growing mixed vegetable…

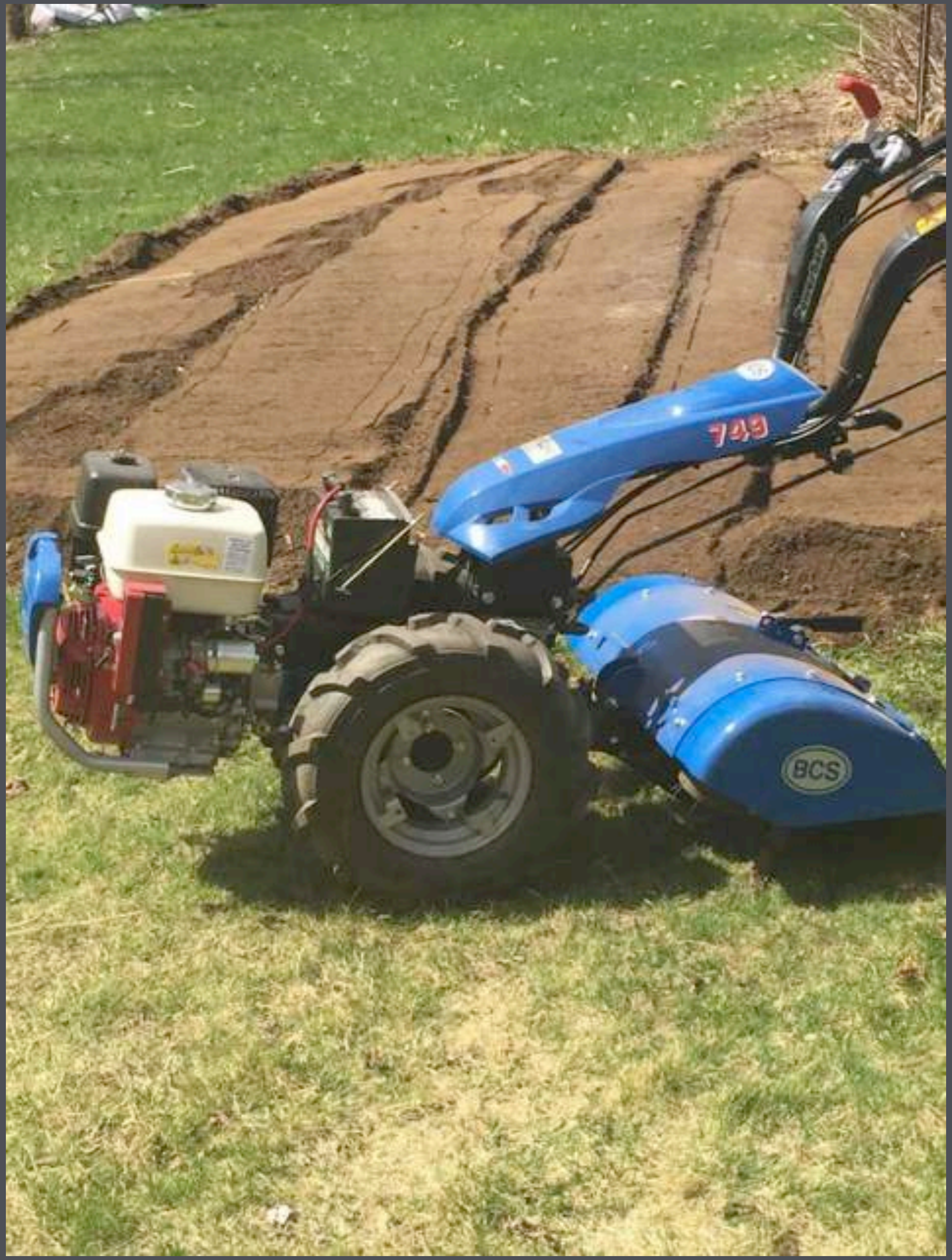
The limited factor is **time** and **economics**

What is No-Till?

- Farming without machines such plow, rototillers or spaders that turn over the ground
- follows another rule (often ignored) disturb your soil as little as possible (or not at all)
- Good but not enough in its self

Most market gardeners believe it is essential to till the soil. Why ?

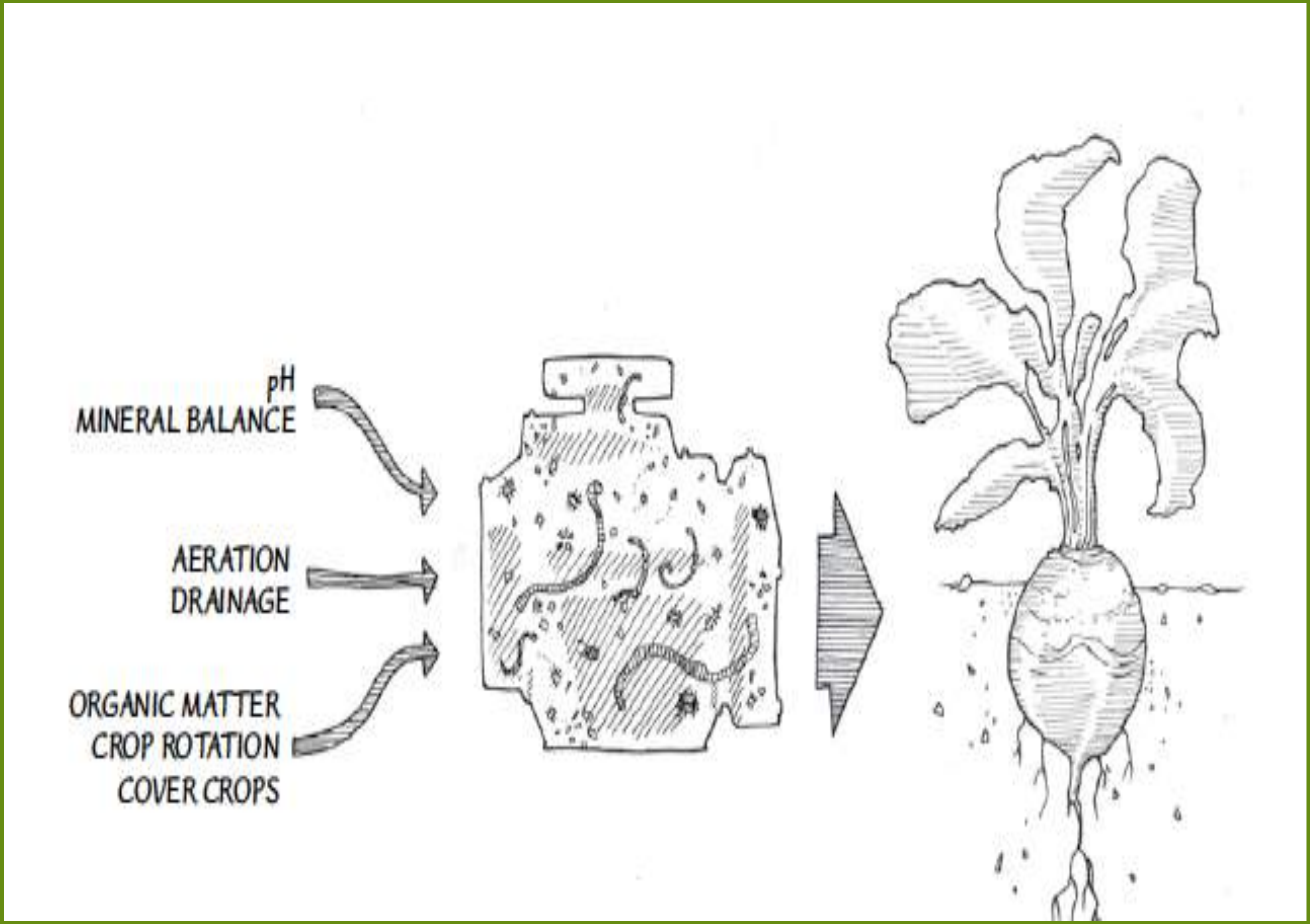
- They think roots need loses soil to penetrate and grow in it
- Its easy to prepare clean seed beds. F*&# convenient



$\uparrow$ Tillage α $\downarrow$ Organic matter

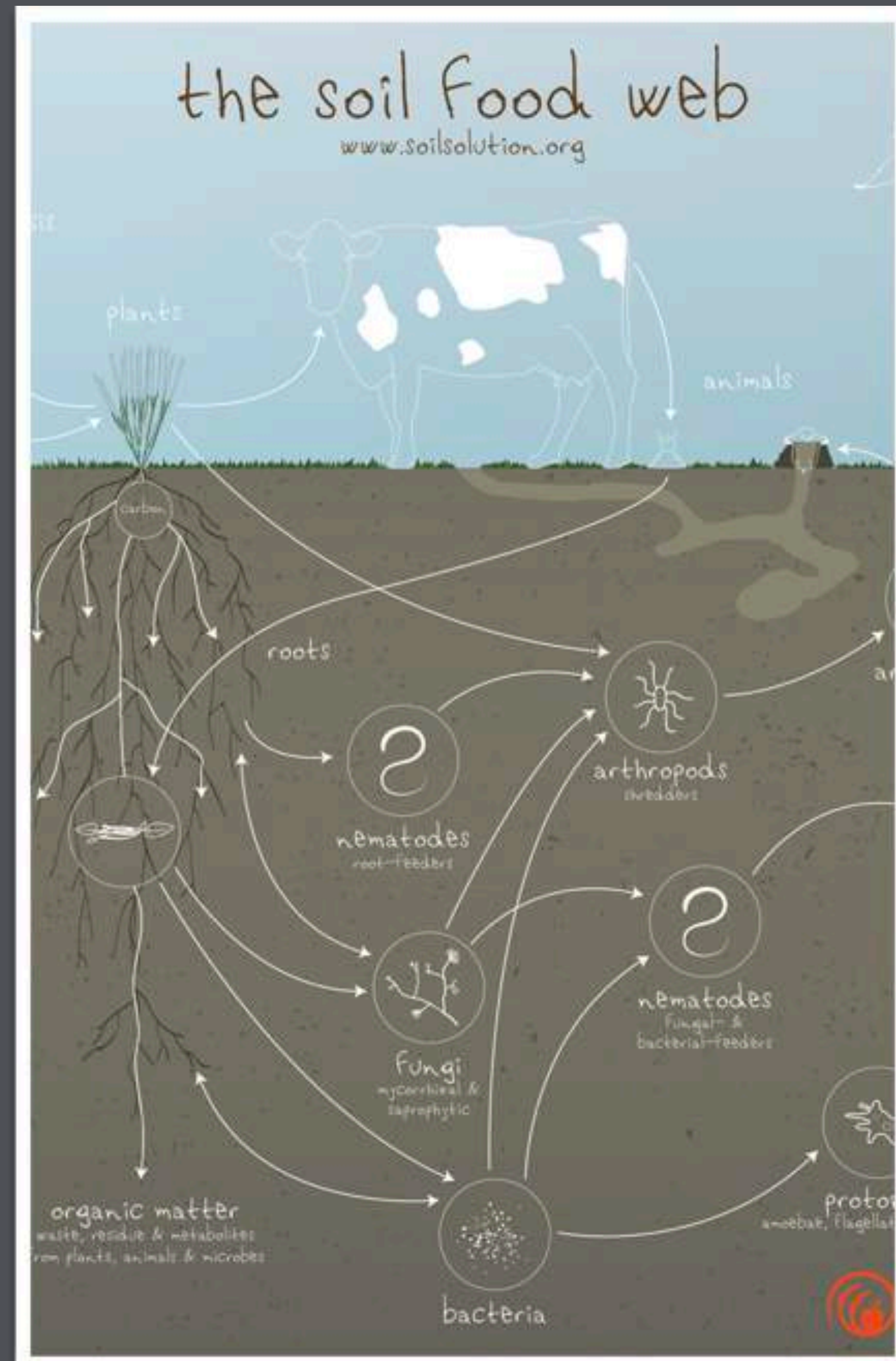
No-till vs Working with Living Soils

Under the right conditions, the living web does all the work



Short Biology Primer

Organic matter is the driven factor of fertility.

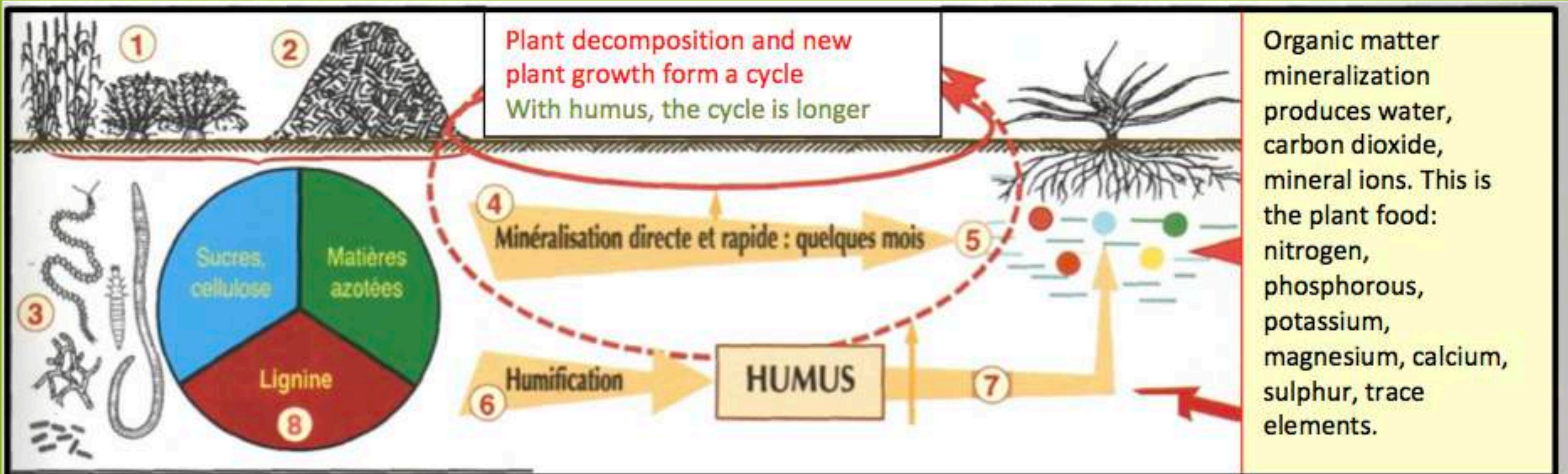


How is humus created and what are soils' "organic materials"?

- Vegetable residues are deposited on soil or ground up and stacked in compost In the soil, the roots are constantly chewed up by insects and earthworms, fungi and bacteria

- One part is mineralized directly for plants nitrogen N, phosphorous P, potassium K, sulphur S, calcium Ca, magnesium Mg, and many elements in low concentrations, known as trace elements (boron, cadmium, chrome, copper, nickel, lead, zinc).

- Another part is transformed into humus which will act as an agglutinate for the soil particles, called humification. And, very slowly, this humus will mineralize and produce the same plant foods.



The main raw material in humus is lignin (8), a component of old plant tissues such as wood. To this lignin, other substances are added: sugars, cellulose and nitrogenous matter, which are found in leaves and straw. These, however, do not contain lignin.

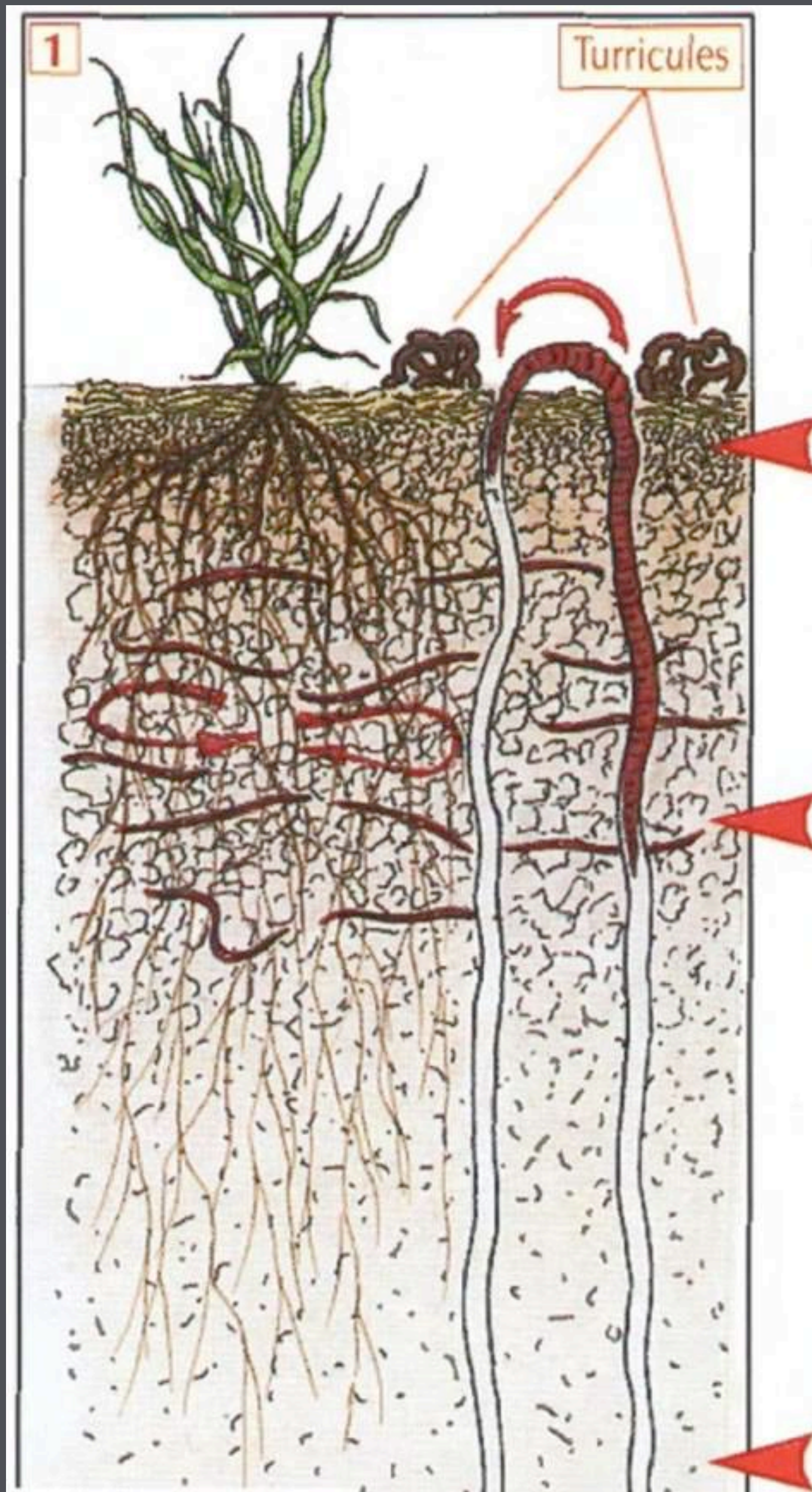
That is why GREEN WASTE COMPOSTS, also called PLANT COMPOSTS or GREEN COMPOSTS
are best at producing humus, since they bind together:

The lignin from wood trimmings and the cellulose from straw and dry grasses

Sugars from leaves and lawn clippings, main food for the bacteria

Nitrogenous matter from leaves, lawn, possible manures...

Organic matter feeds the soil just as it feeds the plants. As growers, its are main input.



On the surface, the EPIGEAL fauna (cuts up plant residues, digests them and returns fecal pellets. Here we find small worms and, especially, a myriad of insects, mites, centipedes and springtails. They facilitate the colonization of plant debris by fungus.

Beneath, the HYPOGEAL fauna is comprised of small worms which build mostly horizontal tunnels. They feed on dead roots from prior crops, leaving numerous channels, thereby aerating the soil and creating pathways for new roots to penetrate through. Their excrements become prey for bacteria which then turn them into food for the crops.

And from top to bottom, ANECIC worms are the night crawlers which ingest plant debris on the surface, then mix it in their digestive tract with the fine soil they collect, and return the mixture into the cavities they've laid out and onto the surface in the shape of worm casts. This is the origin of "clay-humus complex", an intimate mixture of clay and humus.

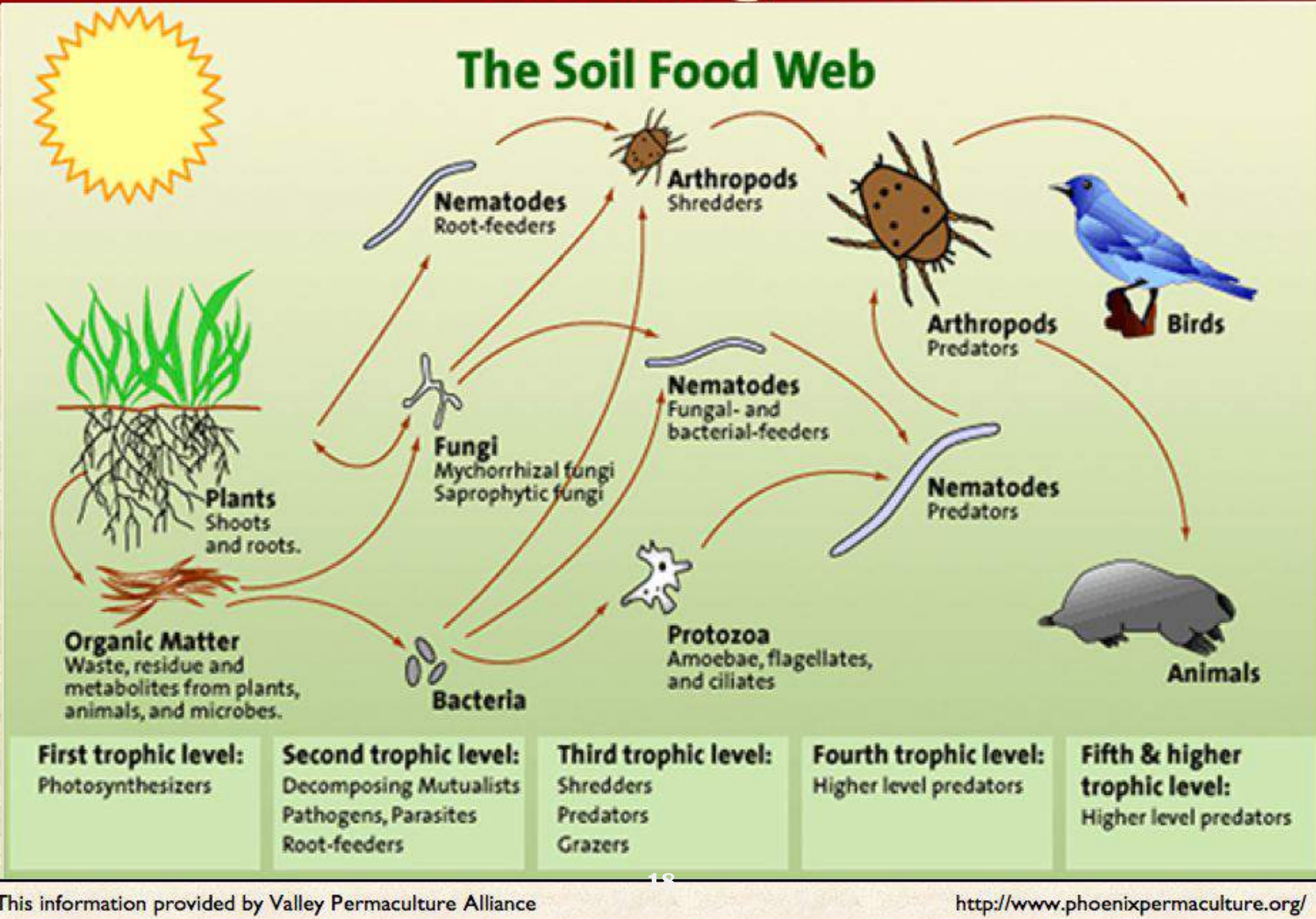
Working with living soils: challenges for veggie production

- Easier to start a crop with clean slate
- Margins are slim, time is scarce
- Tilling has some advantages:

- **We don't work the Forest**, , yet the leaves, twigs and dead branches decompose and feed the deep roots of trees. The soil fauna is then able to "work" the soil and incorporate all plant residues, alive or dead.

- **Open up PRAIRIE soil** using a pitchfork to see how the "hairs" of grasses (ray grass, cocksfoot, ...) have broken the soil down into fine lumps, in other words "built" the soil and given it a "crumbly and aerated structure" And, at the same time, a "STABLE structure" making them resistant to crushing by rain and by other sources of compression, to erosion, to freezing and to crusting.

Soil Building Basics



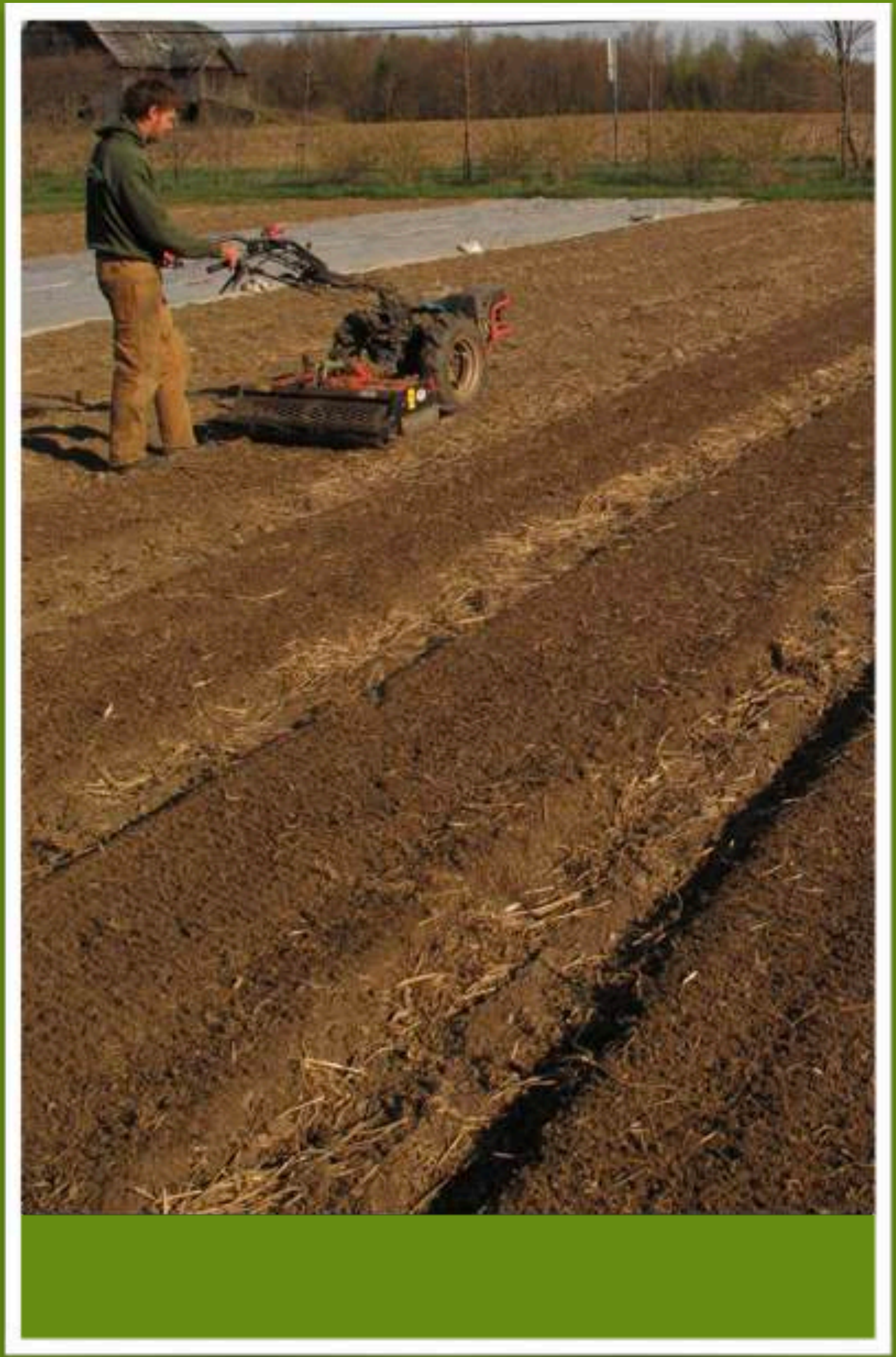
- **In a MARKET GARDEN**, repeated supply of manure, crop tops, composts, used alternately with green manure, allows the soil fauna to structure the soil, to "build" it: the clods break up easily and uncover a whole network of channels, pores and tunnels.

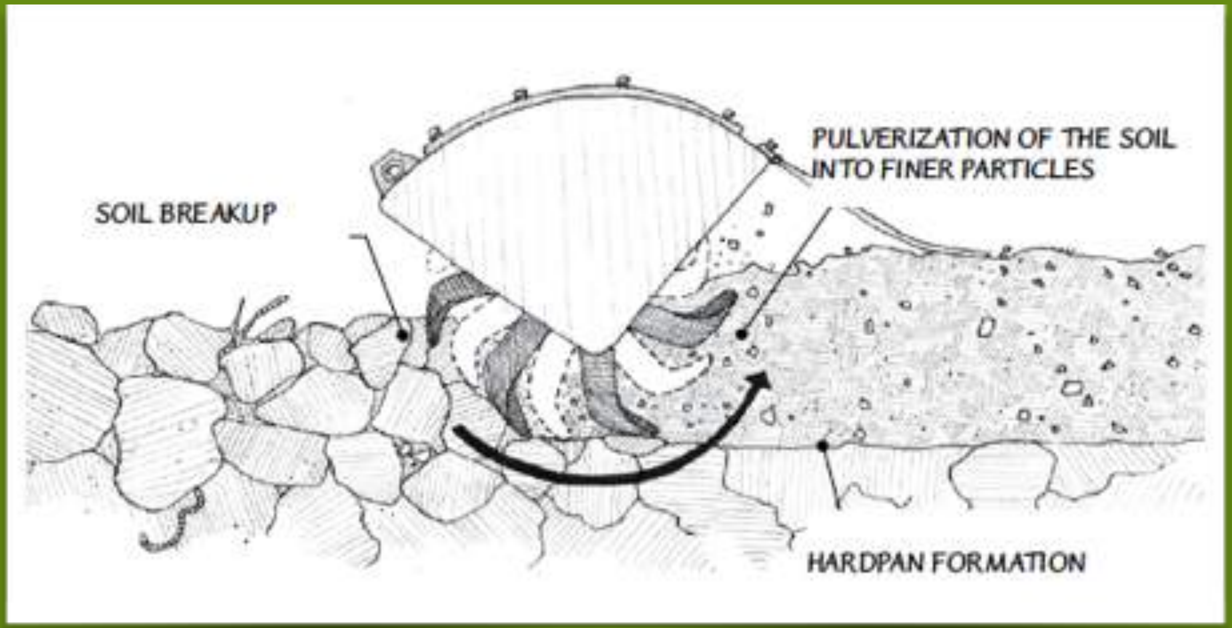
- To upset this construction, this fauna habitat, by turning it or tilling it, under the pretense of destroying a few weeds, is therefore as unnecessary as it is harmful.

Working with Living Soils: Different Strategies

- Moving away from rototiling
- Rely on occultation to clean our beds
- Use tools, like a broad forks to help at initial stage of the process
- Adopted a permanent bed system where the soil is never compacted, not heavily disturbed by plowing
- Use ramial wood chips in our fertility program
- Use cover crops and the power of root systems to feed our soil web
- Cropping directly onto compost
- Cover the ground as fast, and as much as practical
- Applying mycorrhizal inoculants at germination (in the potting mix)
- Planting into cover crop mulch
- Using compost tea in a systematize way

- Moving away from rototiling





« Tilling the soil is the equivalent of an earthquake, hurricane, tornado, and forest fire occurring simultaneously to the world of soil organisms. »

–Don Tyler, a USDA conservation specialist
Appleseed

- Relying on occultation to clean our beds

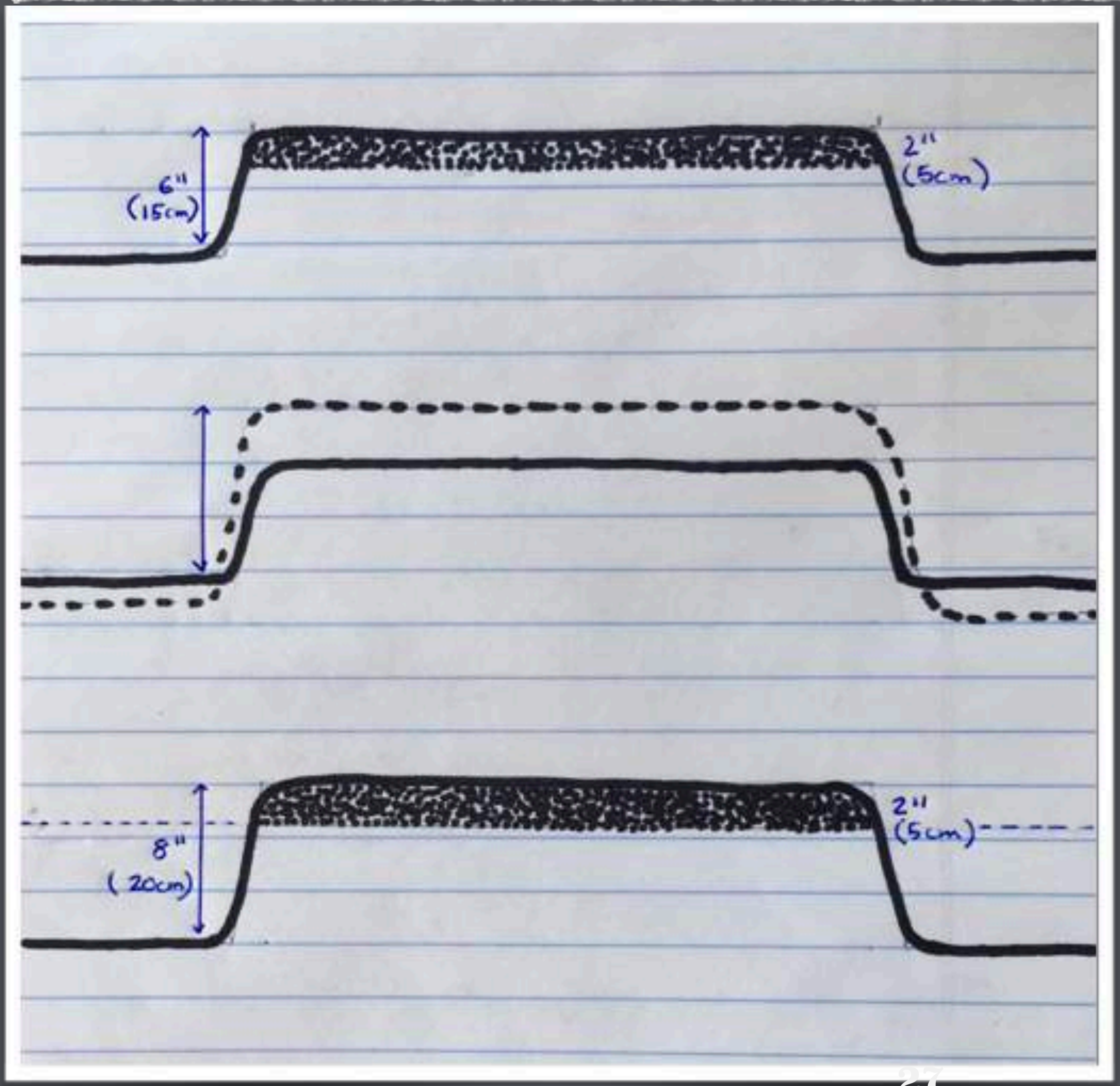
- UV treated black village tarps

- plastic mulch" and they are extremely effective–at suppressing weeds, holding in moisture, and nurturing soil microbes

- Use tools, like a broad forks to help at initial stage of the process

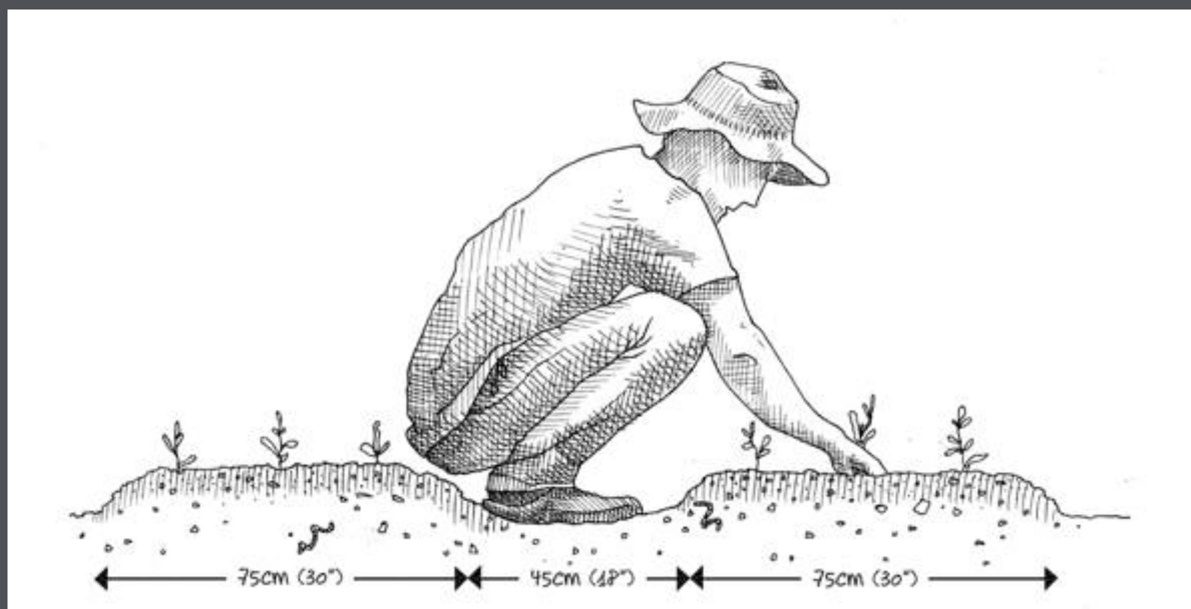

www.themarketgardener.com

Cheating the no-till.



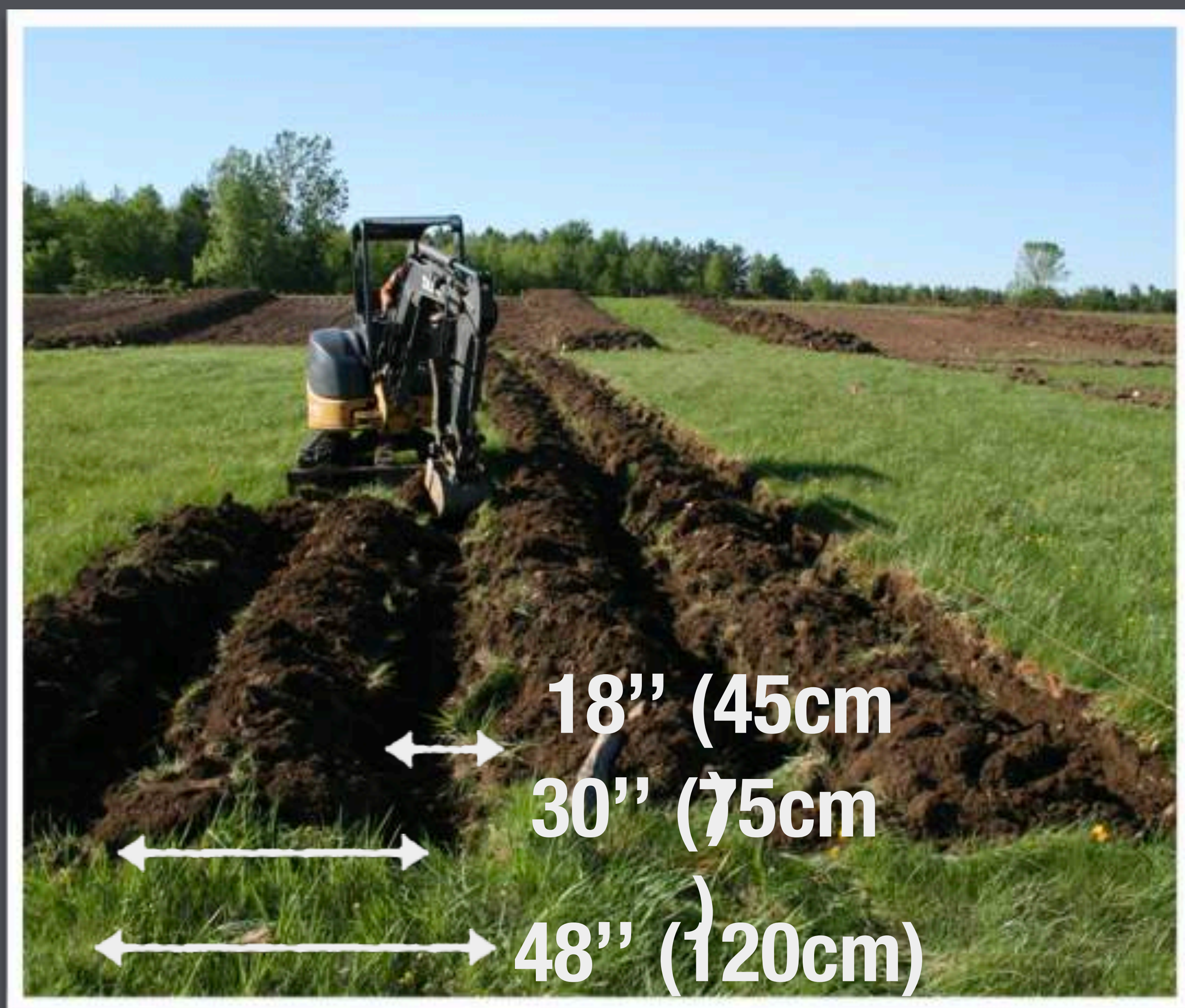
- Keep upper soil intact, by hilling our pathways and working this new 2 inch

Permabeds 2.0



Why go trough the trouble of making new beds every year ?

How to start your permabeds (when the future garden is currently lawn)



- The fast way is to rent a mini-excavator +/-48′ center to center. Inside tracks must have +/-32″.
- Dig a 10″ trench every 4 feet which you flip onto the 30″ bed. The bucket of the excavator needs to be 18″ which is a standard but doubled check.
- Making 10 beds takes about 2 hours

For perennials with deep root systems (bindweed, thistles) we manage by covering the bed with a black film (photo 6) and leaving it in place for many months, even up to a year or more

- If the soil is clayey, ochre-coloured (therefore low in humus), it will take longer for the mulch and it’s fauna to structure the soil. It might be necessary to work the soil in the beginning;
- Its time to invest in organic matter
- 2’’ X 30’’ X 1200’’ = 3 Cubic yard @ 45$ = 135$ per bed

What to do if we only get started in the spring

- Mark your beds using lines
- Add 1 or 2 inch of compost directly unto to grass
- Cover this layer of compost with thick layers of mulch, either before or after transplanting the vegetables (for example, lettuce).
- Trap is possible for at least 2 to 3 weaks, after which you should be able to transplant into. If direct sown, rove straw and seed unto

Ramial Wood Chips (RWC)
A Canadian rediscovery joining the millennial action of forests

- Wood lignin is the main ingredient in stable humus, the type of humus which retains water and nutritional elements.
- When incorporating the wood chips, its the fungi, not the bacteria that compost it.
- The white mold it creates, is the “forage” of a diversified fauna comprised of worms, nematodes, insects, mites, springtails, etc… Their excrements then become the prey for bacteria which mineralize them to benefit the plants. The cycle is complete

In any case, incorporating wood into the soil can't be bad, seeing as the forest has always been doing so….

What is RWC exactly

- Ramial chipped wood (RCW), also called BRF (from the french name, bois raméal fragmenté, "chipped branch-wood"), is made solely from small to medium-sized branches of hard woods

- It can also be the twigs and branches of woody shrubs given that its no more then 2′′ (7 cm) in diameter.

- It’s processed into small pieces by chipping the branches, ideally in the spring when they are filled with nutrients Thus, it is higher in nutrients and is an effective promoter of the growth of soil fungi and of soil-building in general.

It may be laid on top of the soil (as in mulching), mixed into it (as a green manure), or composted first and then applied. We do the latter. WHY ?

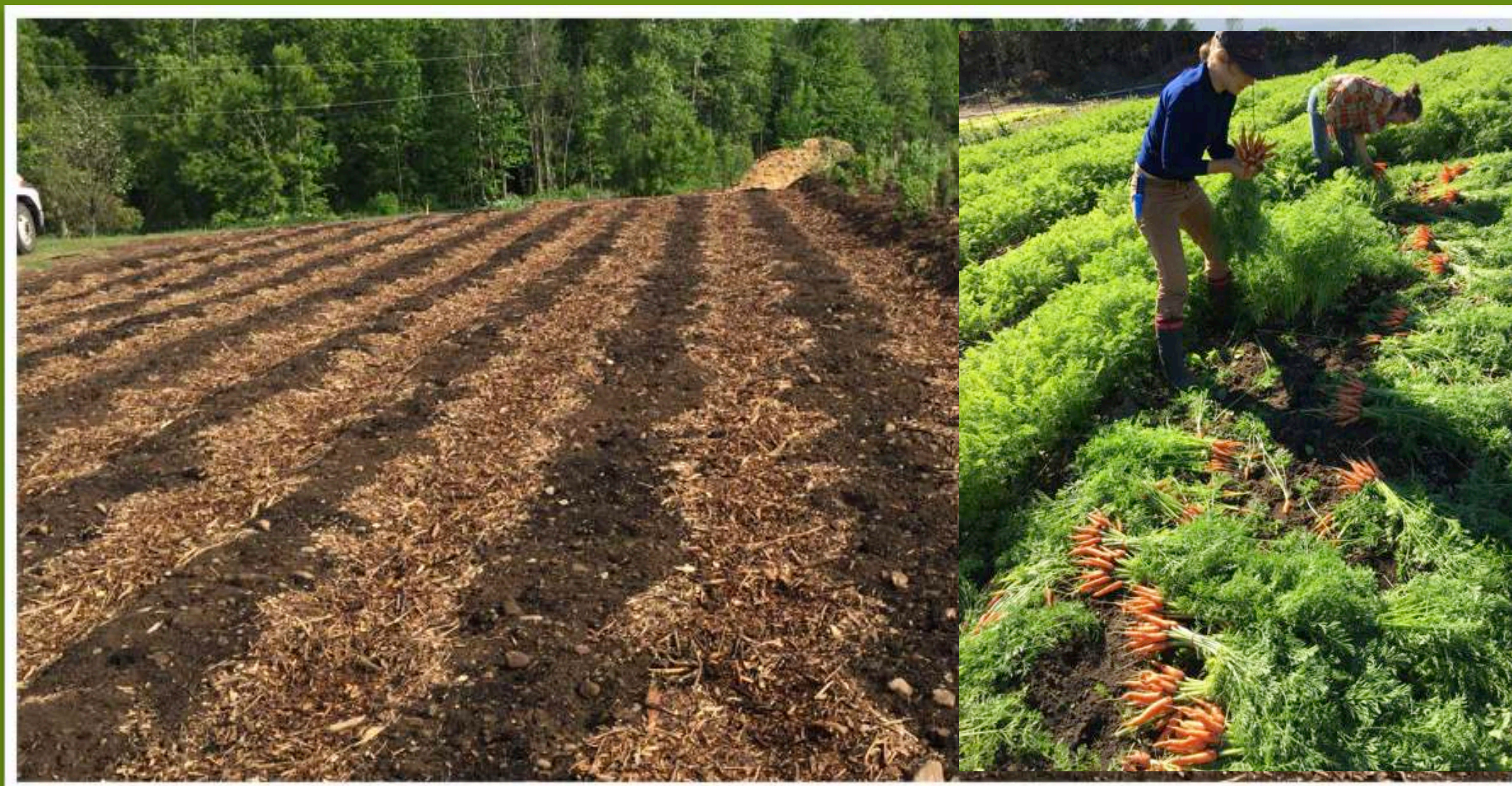
ITS NEEDS TO BE Digested without Veggies otherwise it take the N for your soil

RWC on permabeds: How to

- spread it at least 6 months before incorporating with the plantings
- Use the lowered pathways as the composter (more water, wont create nitrogen loss)
- RWC gets incorporated the next year when beds are raised
- OR spread directly unto beds, but into a legume cover crop. It will make for a nitrogen-fixing humus

- This approach might exempt you from working the soil, but it doesn't exempt you from having to haul the mulch

Our results on different crops aren't significant at the second year, BUT



Fungal activity is obvious with lots of mycelium present,

It is a slow building process, BUT a very powerful one for Soil fauna biodiversity is maximal, forming a food chain connecting wood-rotting fungi, earthworms, insects, mites and springtails.

THEORY GOES that It is a slow building process, BUT a very powerful one for Soil fauna biodiversity is maximal, forming a food chain connecting wood-rotting fungi, earthworms, insects, mites and springtails.

Introducing spore inoculum into the potting soil

- Seedling that are potted up dont need to be inoculated again.

What is green manure and why grow it in the market garden?

- “Green manure” or “ground cover” is a crop destined first and foremost to improving the soil, its fauna and the subsequent crops.
- Think of them as your soil feeders.
- Fall and winter is just perfect for them

- Green manure rapidly covers the ground, protecting the soil from surface sealing, against runoff and the erosion of sloping gardens. Protection, regardless of the season.

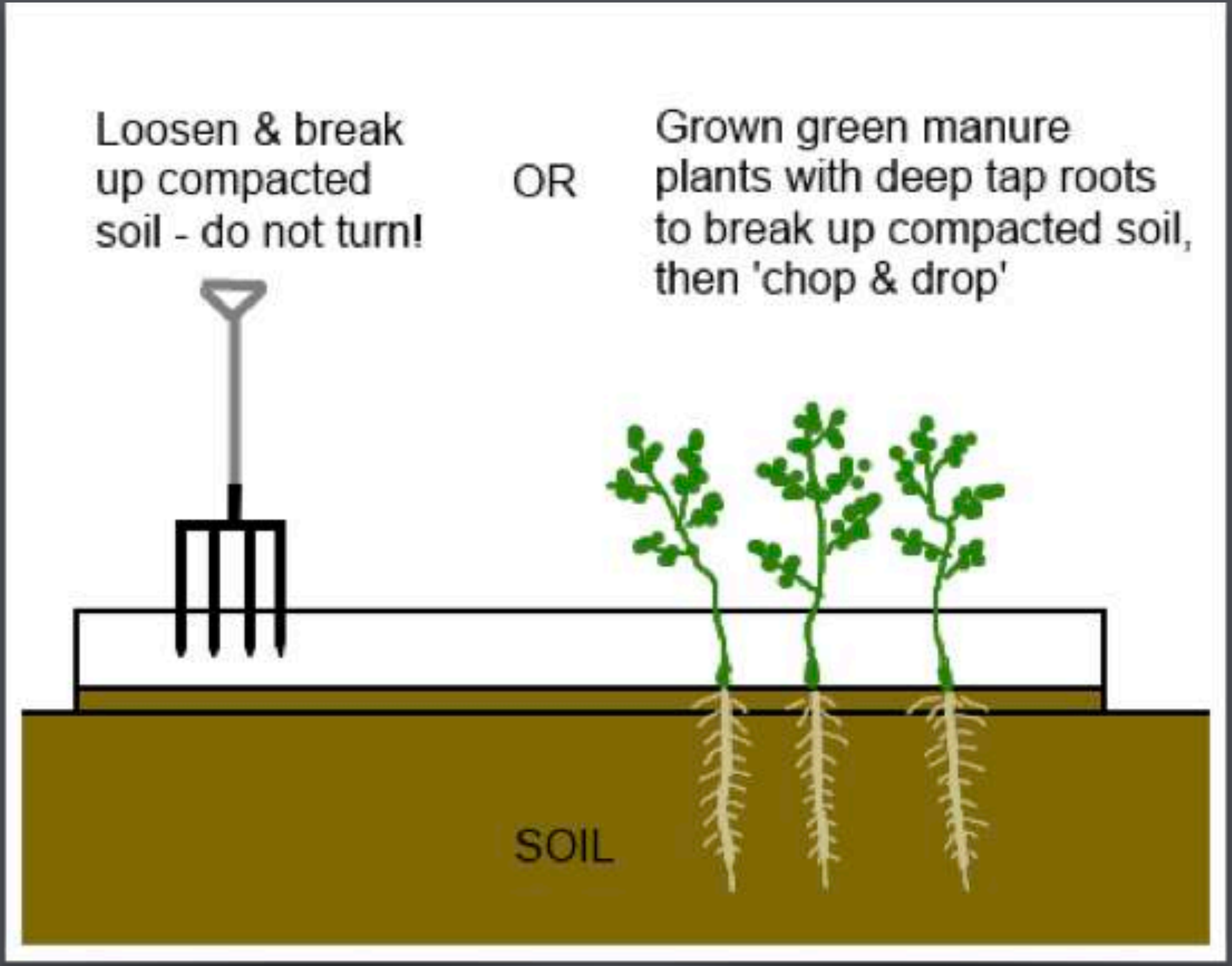
- Their roots create channels throughout the soil, on the surface as well as in depth. Their death leaves behind, in these channels, humus which agglutinates the earthy aggregates

- The decomposition of their plant mass, above and underground, helps the soil life to develop: worm and insect fauna, fungi and bacteria microflora.

- Their roots secrete enzymes that attack the insoluble soil supplies (phosphorous, potassium...). The decomposition of green manure releases these elements for the crops.

- If they are legumes (clover, vetch, peas...), their root nodules transform the air's nitrogen into proteins which will, in turn, give off nitrates that will feed the subsequent crops.

Let biology do the job



- The rootlets create channels in the soil and surround the earthly aggregates
- Aggregates: grains of sand and loam covered in clay and humus
- The dead roots will turn into humus, the dement for aggregates

The end of green manure: organic incorporation into the soil

- Most farmers bury green manures into their soil . Serious mistake ?
- Why not let them decompose on the surface. It’s the soil fauna that will incorporate it into the soil
- To enable this, we bury them UNTO the soil and cover it with silage tarp

Cropping directly onto mulch compost

- Replace working the soil
- The compost is our equivalent of rototillers and spaders.
- Prevents weeds
- Fertilize

Planting ON compost mulch consists in covering well-leveled soil with a layer of compost as a support for seeds.

- Seedings made possible by the specific properties of composts made in facilities:
- Total absence of foreign seeds
- Inexpensive in bulk
- Widely available

Why “bought” composts?

- “Ripe” Their fermentation is complete. It’s therefore possible to seed directly on it. Compost in the process of fermenting would be detrimental to the emergence of the seeds since it contains agents that inhibit germination, in particular ammonia.
- “Poor” Their nitrogen content is low (about 1%) compared to manure composts. A characteristic which confirms that, in the garden, these plant waste composts don’t present a risk for overdosing or for washing away.
- “Decontaminated” Their fermentation temperature and the duration of their composting-maturation are such that they’ve destroyed all seeds and disease-causing germs. A quality that garden composts don't have, even when made carefully.

- Spread ON the soil using the back of the rake: The weeds seeds, deprived of light, wont germinate
- The use of “bought compost” makes this possible made possible thanks to the 2 qualities described earlier :
- This compost is RIPE, meaning it won’t hinder germination
- is “DECONTAMINATED”, meaning it doesn’t contain any foreign seeds, all of them having been destroyed by the lengthy fermentation process

Ideal for densely seeded crops Like those seeded with the 6 row seeder (12 rows on 30′) and harvested with the greens harvester


Maintaining is critical. We dot it by watering daily AND-0r covering with a floating row covers limits evaporation in summer

is scale-uppable ?

Cover the soil as much as possible

- Use landscape fabric for crops 50 days or more
- Use insect nets or row covers regularly. It counts
- Intensive spacing for short crops also act a ground cover
- Use rye straw for long season crops which like cooler soil

Working with living soils
Next steps

- Integrate compost tea or other micro-organism solution (Korean Natural Farming, Bio-Dynamic, etc) into our operational procedure

- Integrate grass clipping in our systems (higher in N and more potent then Alfalfa meal)

- Use mowed cover crops systematically for certain transplanted veggies (Broccoli, cauliflower, etc)


WARNING
GEOTEA
Compost Tea Machines
Sweetwater

Preemptively spraying for health (as opposed to symptoms) changes the disease challenges

How about that?

Is it efficient ?

Food not lawn

REMEMBER… WORKING WITH LIVING SOILS IS A DIRECTION, NOT A DOGMA.

CHANGES CAN BE DONE INCREMENTALLY OVER SOME YEARS